

MATH3900: Optimization for AI and DA

Introduction

Prof. Lexiao Lai
Department of Mathematics
The University of Hong Kong

Semester 1, 2026–2027



► Weekly meetings:

	Time	Venue
Lecture	Tue 09:00–09:50	KK101
	Thu 10:00–11:50	KK101
Tutorial	Fri 12:00–12:50	LE2

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- Any use of artificial intelligence (AI) tools must be acknowledged.
- **Academic integrity: submit your own work.**

Regarding course content:

- ▶ MATH3904 (*Introduction to Optimization*) covers content similar to that of MATH3900, so the two courses are mutually exclusive.
- ▶ The main difference is that MATH3900 focuses on optimization theory and algorithms motivated by applications in AI and data analytics, whereas MATH3904 develops a general theory of optimization.

MATH3900 vs MATH3904

Regarding course content:

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Regarding degree and major requirements:

- ▶ Both MATH3900 and MATH3904 can be used to satisfy the major requirements for [Applied AI](#) and [Decision Analytics](#).
- ▶ MATH3900 is **not** a disciplinary core or elective course for **Mathematics** majors or minors.
- ▶ Students in the **Mathematics (Intensive)** major should **not** take MATH3900 because MATH3904 is compulsory for that major.

What is an optimization problem?

- ▶ An optimization problem asks us to choose a decision variable x that makes an objective as small as possible while satisfying the constraints:

$$\min_{x \in C} f(x).$$

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continuous optimization

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- **Decision variable** $x \in \mathbb{R}^d$; **Objective function** $f : \mathbb{R}^d \rightarrow \mathbb{R}$.

- **Feasible region:** $C \subset \mathbb{R}^d$

① $C = \{ \|x\| \leq 1 \}$



② $C = \{ g_1(x) = 0, g_2(x) \leq 0 \}$

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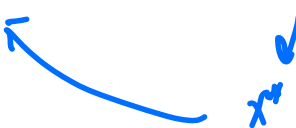
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- ▶ How about maximization problems?
- ▶ A maximization problem can be written as a minimization problem:

$$\max_{x \in C} f(x) \quad \Longleftrightarrow \quad \min_{x \in C} [-f(x)].$$


Optimal solutions

Consider the optimization problem:

$$\min_{x \in C} f(x).$$

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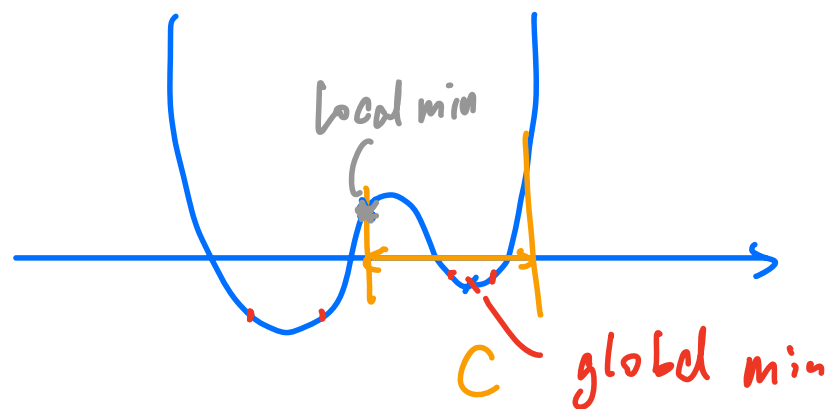
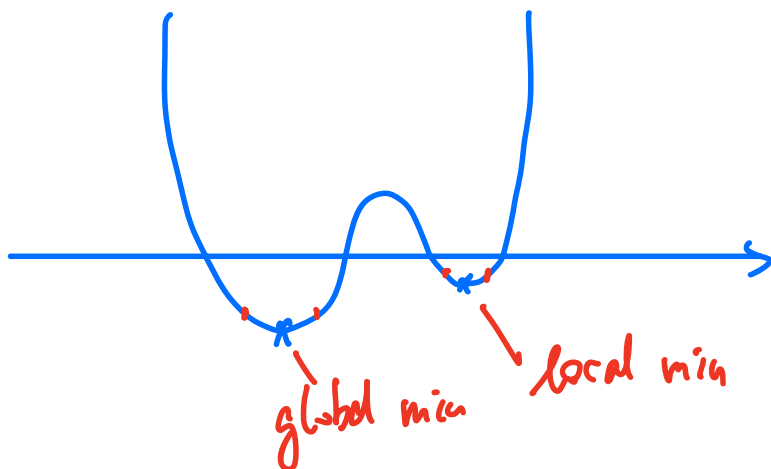
$$\min_{x \in C} f(x).$$

Definition

For $f : C \rightarrow \mathbb{R}$ and $x^* \in C$:

- ▶ x^* is a **global minimizer** if $f(x^*) \leq f(x)$ for every $x \in C$.
- ▶ x^* is a **local minimizer** if there exists $\varepsilon > 0$ such that $f(x^*) \leq f(x)$ for all $x \in C$ with Euclidean distance $\|x - x^*\| \leq \varepsilon$.

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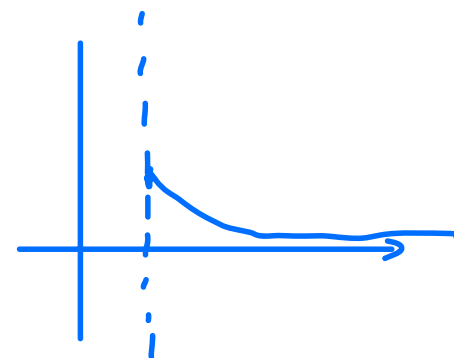
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- ▶ The notation $\arg \min_{x \in C} f(x)$ denotes the set of global minimizers.
- ▶ The **optimal value** is $f^* := \inf_{x \in C} f(x)$. *Largest val y s.t. $y \leq f(x), \forall x \in C$*
- ▶ If x^* is a global minimizer, then $f(x^*) = f^*$.

$$C = [1, \infty) \quad f(x) = \frac{1}{x}$$

$$f^* = 0$$



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- ▶ The **optimal value** is $f^* := \inf_{x \in C} f(x)$.
- ▶ If x^* is a global minimizer, then $f(x^*) = f^*$.
- ▶ Is the optimal value always obtained at some $x \in C$?

When does a minimizer exist?

Theorem: Weierstrass theorem

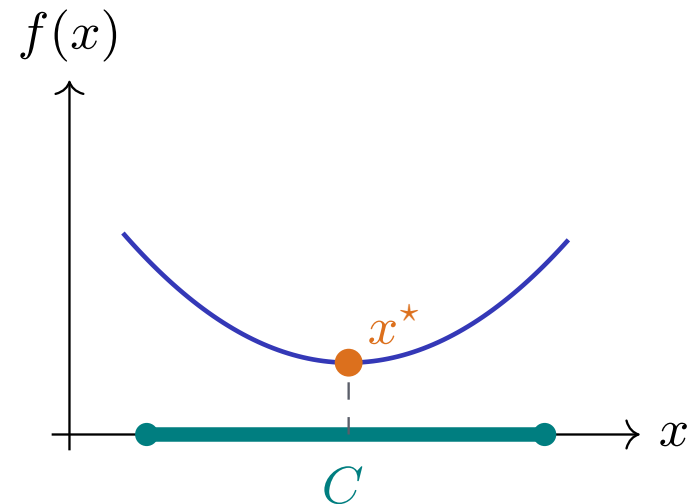
If $C \subset \mathbb{R}^d$ is nonempty and compact and $f : C \rightarrow \mathbb{R}$ is continuous, then f has at least one global minimizer on C .

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If $C \subset \mathbb{R}^d$ is nonempty and compact and $f : C \rightarrow \mathbb{R}$ is continuous, then f has at least one global minimizer on C .

- In $\mathbb{R}^d$, a set is compact exactly when it is closed and bounded.

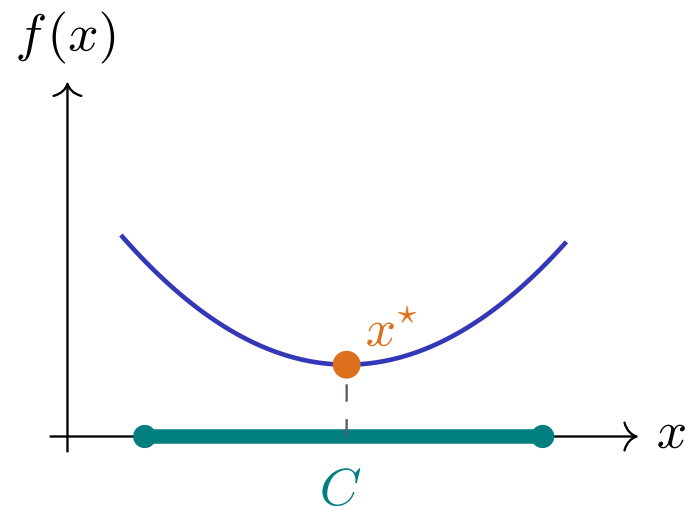


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- ▶ In $\mathbb{R}^d$, a set is compact exactly when it is closed and bounded.
- ▶ Closedness: The set contains its boundary points.

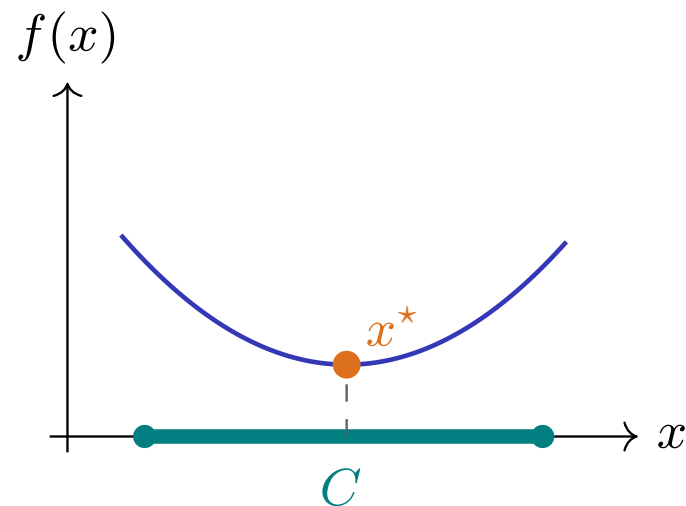


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- ▶ In $\mathbb{R}^d$, a set is compact exactly when it is closed and bounded.
- ▶ Closedness: The set contains its boundary points.
- ▶ Boundedness: The set is contained in a ball. $\exists r > 0$, s.t.
 $\forall x \in C, \|x\| \leq r$

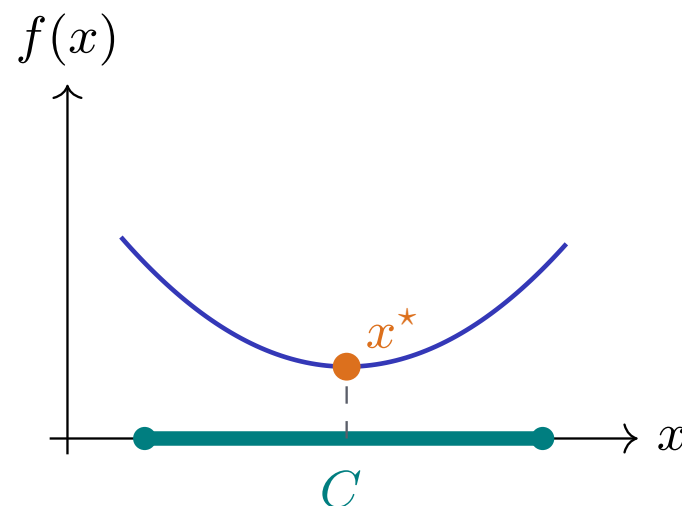


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- ▶ Are these conditions necessary?

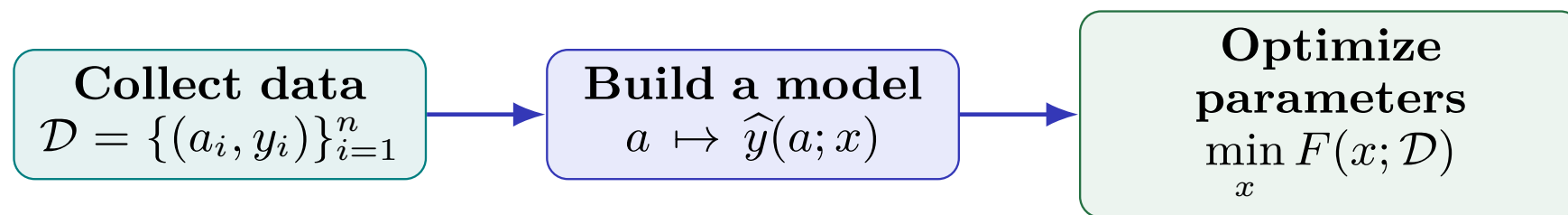


Optimization problems in AI and Data Science

- ▶ Many AI and data-science tasks can be formulated as optimization problems.

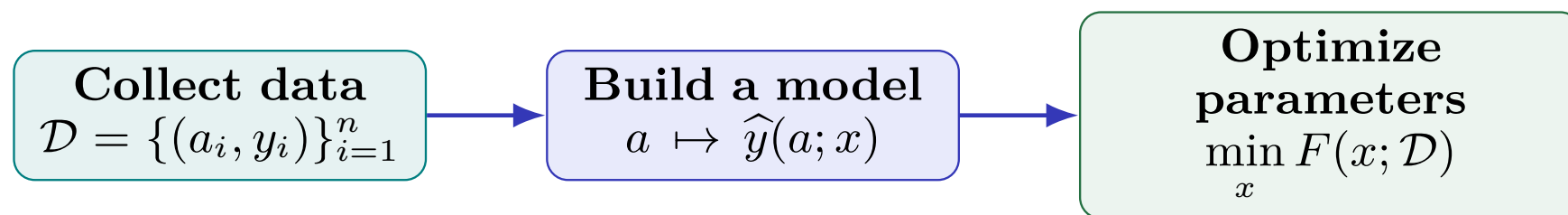
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Optimization problems in AI and Data Science

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Task	Optimization problem
Linear regression	$\min_x (2n)^{-1} \sum_i (a_i^T x - y_i)^2$
Binary classification	$\min_x n^{-1} \sum_i \log(1 + e^{-y_i a_i^T x})$
Hard-margin SVM	$\min_x \frac{1}{2} \ x\ ^2$ subject to $y_i a_i^T x \geq 1$ for all i
Sparse prediction	$\min_x f(x) + \lambda \ x\ _1$
Neural-network training	$\min_x n^{-1} \sum_i \mathcal{L}(h(a_i; x), y_i)$

Example: Linear Regression

- ▶ **Task:** predict a numerical response $y \in \mathbb{R}$ from features $a \in \mathbb{R}^d$.
GDP *GDP previous year, population*

Example: Linear Regression

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$$\hat{y} = a^T x. = \sum_{i=1}^d a_i x_i$$

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- ▶ This optimization problem has a closed-form solution; we will return to it later.

Example: Logistic Regression

- **Task:** predict a binary class $y \in \{-1, 1\}$ from features $a \in \mathbb{R}^d$.
- cat*
↓
dog
↓
images
↓

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- ▶ We choose x by solving x_i

$$\min_{x \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^n \log \left(1 + e^{-y_i a_i^T x} \right).$$

Handwritten notes: x_i (above the sum), x_i (above the log), $\log(1 + e^{-f_i})$ (next to the log), and a downward arrow.

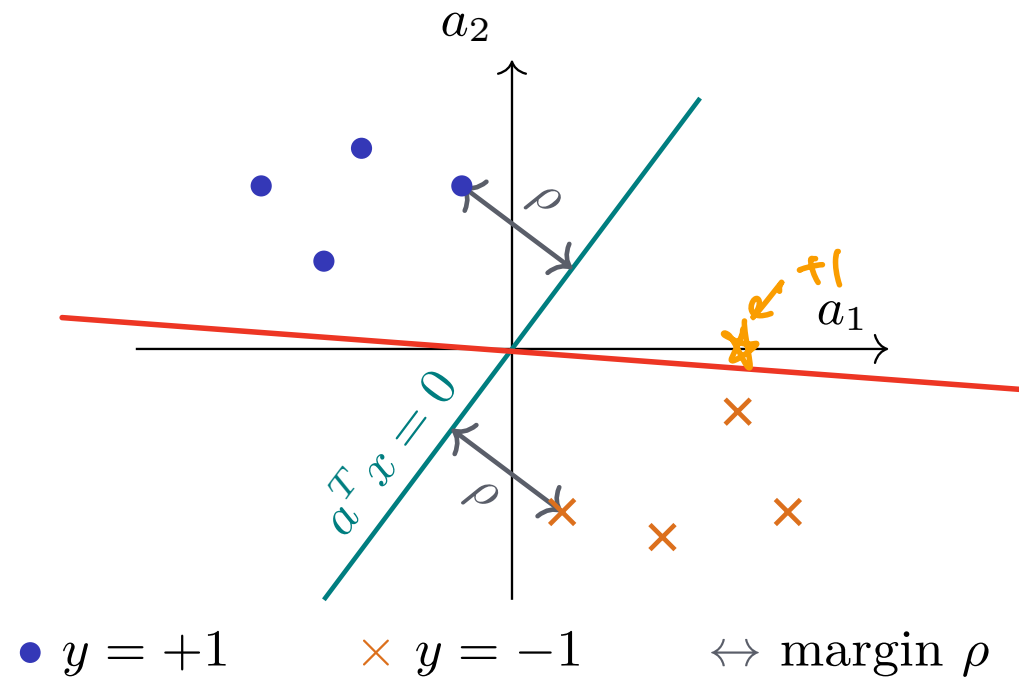
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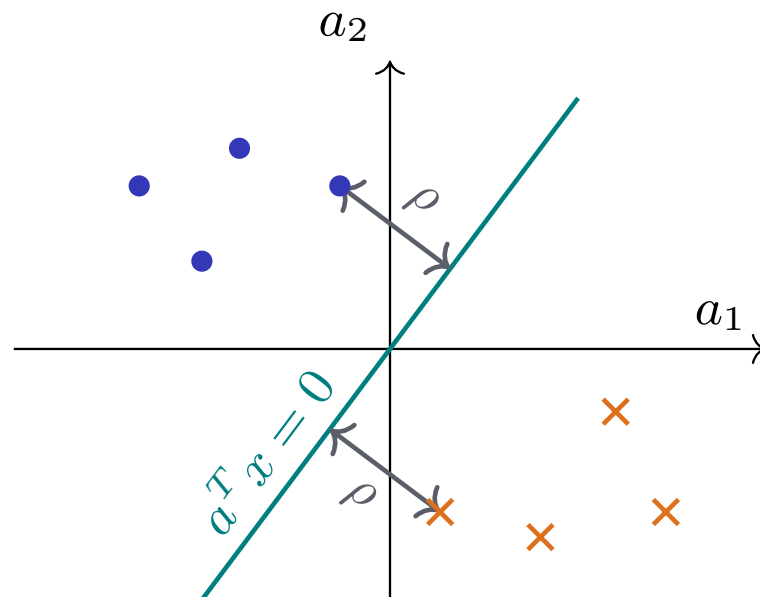
$$\min_{x \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^n \log \left(1 + e^{-y_i a_i^T x} \right).$$

- ▶ This problem does not have a closed-form formula.

Example: Hard-Margin SVM



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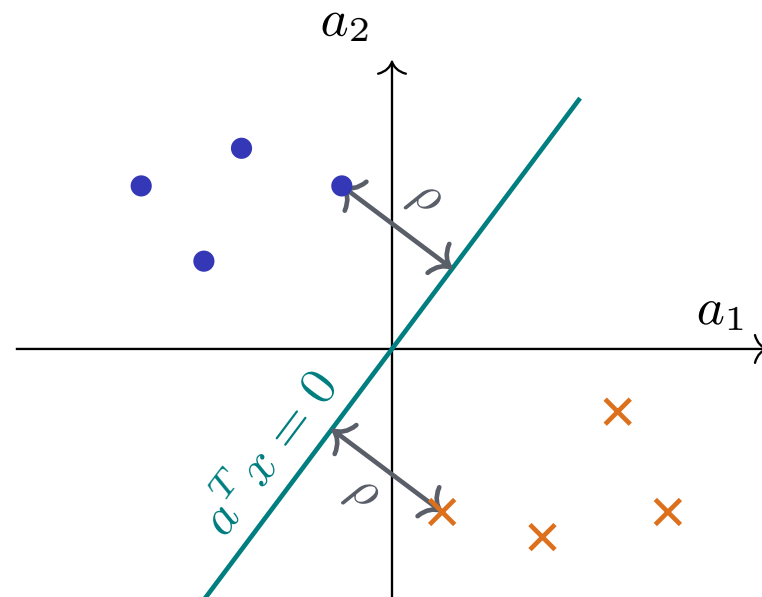


• $y = +1$ × $y = -1$ $\leftrightarrow$ margin ρ

- **Task:** for linearly separable binary data, find the widest-margin classifier by solving

$$\begin{aligned} \min_{x \in \mathbb{R}^d} \quad & \frac{1}{2} \|x\|^2 \\ \text{s.t.} \quad & y_i a_i^T x \geq 1, \quad i = 1, \dots, n. \end{aligned}$$

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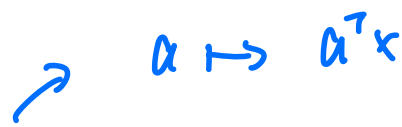
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- This is a **constrained** optimization problem.

Example: Sparse linear regression with Lasso

- **Task:** fit a linear model using only a small subset of the available features.
-  $a \mapsto a^T x$

$$\min_{x \in \mathbb{R}^d} \frac{1}{2n} \sum_{i=1}^n (a_i^T x - y_i)^2 + \lambda \|x\|_1, \quad \|x\|_1 = \sum_{j=1}^d |x_j|.$$

Example: Sparse linear regression with Lasso

$$a \mapsto a^T x = \sum_{i=1}^d x_i a_i$$

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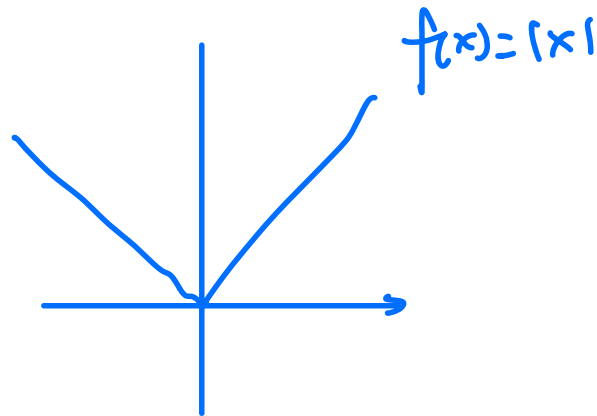
- The parameter $\lambda \geq 0$ balances data fit and sparsity by encouraging some coefficients x_j to be exactly zero.

Example: Sparse linear regression with Lasso

- **Task:** fit a linear model using only a small subset of the available features.

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- ▶ It is difficult to even evaluate F in such large-scale models.
- ▶ Solution: **stochastic optimization**, which uses a small subset (mini-batch) of the data at each time.

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- **Vectors.** For $u, v \in \mathbb{R}^q$ and $\alpha \in \mathbb{R}$, work coordinate by coordinate:
 $(u + v)_i = u_i + v_i$ and $(\alpha u)_i = \alpha u_i$. For example,

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- ▶ **Matrices.** For $M, N \in \mathbb{R}^{p \times q}$, addition and scalar multiplication are also entrywise: $(M + N)_{ij} = M_{ij} + N_{ij}$ and $(\alpha M)_{ij} = \alpha M_{ij}$.

- **Matrix–vector multiplication.** If $M \in \mathbb{R}^{p \times q}$ and $x \in \mathbb{R}^q$, then $Mx \in \mathbb{R}^p$. Each row of M produces one output entry. For example,

$$M = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad x = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad \Longrightarrow \quad Mx = \begin{bmatrix} 1 \cdot 3 + 2 \cdot 1 \\ 0 \cdot 3 + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}.$$

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- ▶ When a $p \times q$ matrix multiplies a $q \times r$ matrix, the inner dimensions match and the result has shape $p \times r$.
- ▶ Many practical objectives can be written compactly in matrix–vector form.

Linear Regression: Rewrite the Objective

- Recall the linear-regression objective:

$$\min_{x \in \mathbb{R}^d} \frac{1}{2n} \sum_{i=1}^n (a_i^T x - y_i)^2.$$

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$$A = \begin{bmatrix} a_1^T \\ \vdots \\ a_n^T \end{bmatrix} \in \mathbb{R}^{n \times d}, \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n.$$

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$$Ax = \begin{bmatrix} a_1^T x \\ \vdots \\ a_n^T x \end{bmatrix} \in \mathbb{R}^n.$$

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- ▶ Substitution gives the same problem in matrix form:

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Directional Derivative and Gradient

Definition

For $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and $x, v \in \mathbb{R}^d$, the one-sided directional derivative of f at x along v , when it exists, is

$$f'(x; v) = \lim_{t \downarrow 0} \frac{f(x + tv) - f(x)}{t}.$$

If f is differentiable at x , its gradient is

$$\nabla f(x) = \left(\frac{\partial f(x)}{\partial x_1}, \dots, \frac{\partial f(x)}{\partial x_d} \right)^T, \quad f'(x; v) = \nabla f(x)^T v.$$

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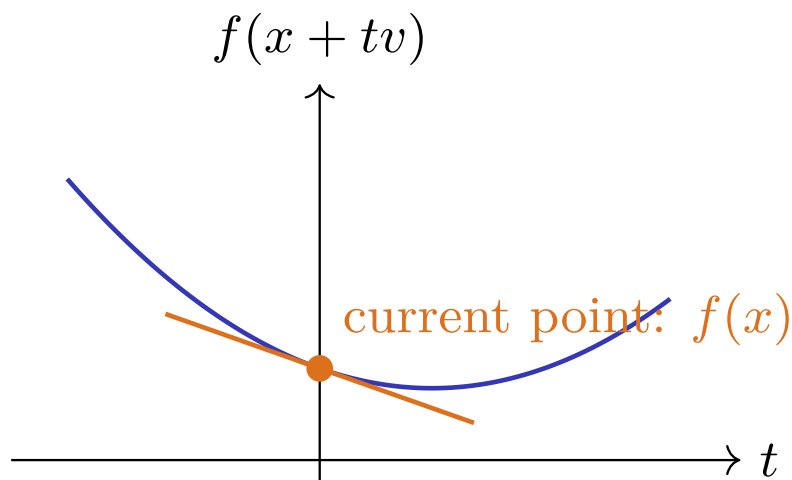
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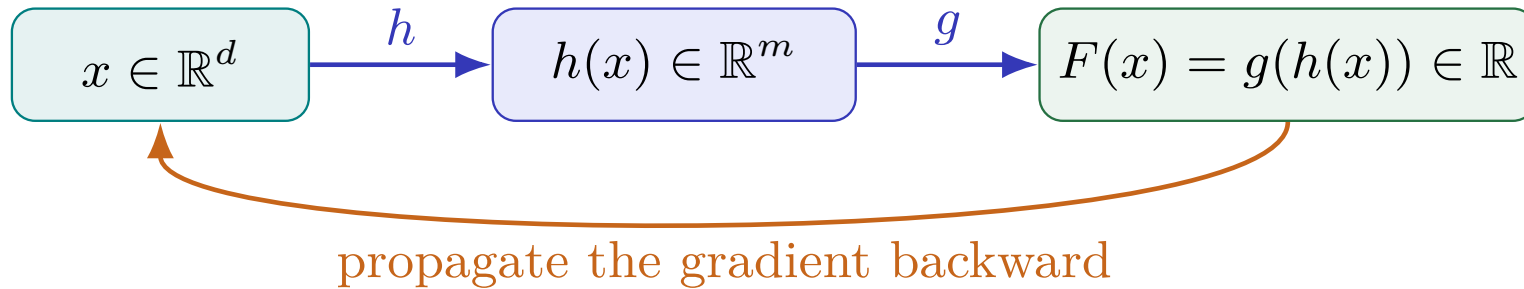
The Chain Rule

- Suppose $F(x) = g(h(x))$, with $h = (h_1, \dots, h_m) : \mathbb{R}^d \rightarrow \mathbb{R}^m$ and $g : \mathbb{R}^m \rightarrow \mathbb{R}$.

$$F : \mathbb{R}^d \rightarrow \mathbb{R}$$

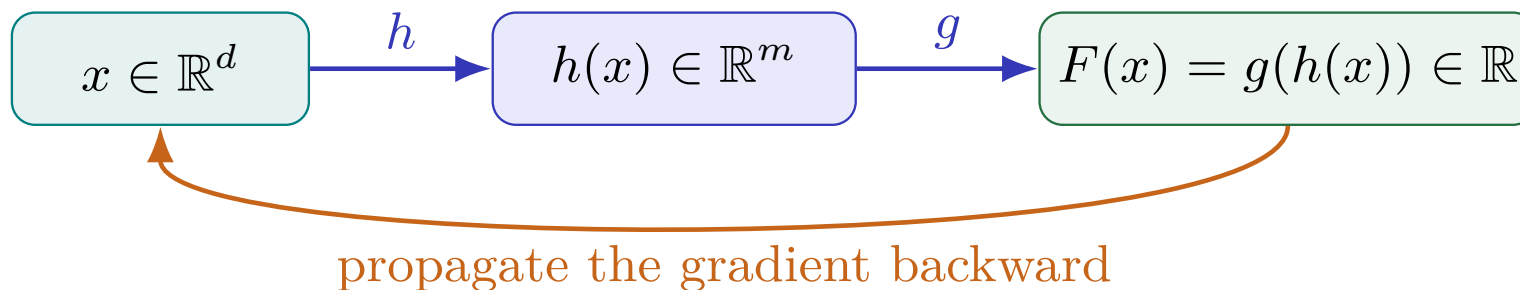
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Theorem

Assume that h is differentiable at x and that g is differentiable at $h(x)$. The Jacobian of the inner map stacks the component gradients as rows:

$$J_h(x) := \begin{bmatrix} \nabla h_1(x)^T \\ \vdots \\ \nabla h_m(x)^T \end{bmatrix} \in \mathbb{R}^{m \times d}.$$

With column gradients, the chain rule can be written in either form:

$$\nabla F(x) = \sum_{j=1}^m [\nabla g(h(x))]_j \nabla h_j(x) = J_h(x)^T \nabla g(h(x)).$$

Linear Regression: Apply the Chain Rule

► Recall $F(x) = (2n)^{-1} \|Ax - y\|^2$. Define the inner and outer maps by

$$h(x) = Ax - y \in \mathbb{R}^n, \quad g: \mathbb{R}^n \rightarrow \mathbb{R}, \quad g(w) = \frac{1}{2n} \|w\|^2.$$

$\mathbb{R}^d \rightarrow \mathbb{R}^n$

The components of h are $h_i(x) = a_i^T x - y_i$, and $F(x) = g(h(x))$.

Find $\nabla F(x) = \underbrace{J_h(x)^T}_{\text{gradient of } h} \nabla g(h(x))$

② $\nabla g(w) = \frac{1}{n} w$

① $\nabla h_i(x) = a_i \leftarrow h_i(x) = \sum_{j=1}^d a_{ij} x_j - y_i$

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$$\begin{aligned} \nabla F(x) &= J_h(x)^T \nabla g(h(x)) \\ &= \frac{1}{n} A^T (Ax - y). \end{aligned}$$

Fundamental theorem of calculus along a line

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If $\phi : [0, 1] \rightarrow \mathbb{R}$ is continuously differentiable, then

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- How to extend such identity to continuously differentiable $f : \mathbb{R}^d \rightarrow \mathbb{R}$?

Fundamental theorem of calculus along a line

Theorem: Fundamental theorem of calculus

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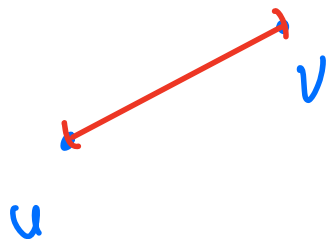
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- ▶ Let $u, v \in \mathbb{R}^d$. Restrict f to the line segment from u to v :

chain rule

$$\phi(t) = f(u + t(v - u)), \quad \phi'(t) = \nabla f(u + t(v - u))^T (v - u).$$

$\phi(0) = f(u) \quad \phi(1) = f(u + v - u) = f(v)$



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- ▶ Applying the above theorem to ϕ gives

$$f(v) - f(u) = \int_0^1 \nabla f(u + t(v - u))^T (v - u) dt.$$

- ▶ Taylor expansions approximate this integral using only derivative at u .

Taylor Expansions

Recall. For $p > 0$, $r(s) = o(\|s\|^p)$ means $r(s)/\|s\|^p \rightarrow 0$ as $s \rightarrow 0$. ~~$s \rightarrow \infty$~~

Theorem

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and fix $x \in \mathbb{R}^d$.

First-order expansion. If f is differentiable at x , then, as $s \rightarrow 0$,

$$f(x + s) = f(x) + \nabla f(x)^T s + o(\|s\|).$$

$$\Leftrightarrow \lim_{s \rightarrow 0} \frac{f(x+s) - f(x) - \nabla f(x)^T s}{\|s\|} = 0$$

Ex. ① $r(s) = 100\|s\|^2$ v.s. $\|s\|$,

$$\frac{r(s)}{\|s\|} = \frac{100\|s\|^2}{\|s\|} = 100\|s\| \rightarrow 0 \text{ as } s \rightarrow 0. \text{ Thus } r(s) = o(\|s\|)$$

② $r(s) = 0.01\|s\|$. Is $r(s) = o(\|s\|)$? $r(s)/\|s\| = 0.01 \not\rightarrow 0$ as $s \rightarrow 0$

$\left\{ \begin{array}{l} \text{If } r(s) = o(\|s\|) \\ \text{then } \forall c > 0 \\ r(s) \leq c\|s\| \text{ for } s \text{ small} \end{array} \right.$

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► Here $\nabla^2 f(x) = [\partial^2 f(x)/(\partial x_i \partial x_j)]_{i,j=1}^d$ is the Hessian.

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- ▶ Here $\nabla^2 f(x) = [\partial^2 f(x)/(\partial x_i \partial x_j)]_{i,j=1}^d$ is the Hessian.
- ▶ Taylor expansions are central to the design and analysis of many optimization algorithms.